

UML & Object-Oriented Modeling — Lab Exercise

Software Engineering (24PCA402) · Unit 06

Objective

Model a Library Management System using UML diagrams that capture both its structure and its behavior.

Task 1 — Use Case Diagram

Identify the actors and the use cases they participate in.

- Actors: Member, Librarian, System Admin
- Use cases: Search Catalog, Borrow Item, Return Item, Pay Fine, Manage Catalog

Task 2 — Class Diagram

Model the core domain classes with attributes, methods, and relationships (association, aggregation, inheritance).

- Classes: Member, Librarian, Book, Loan, Fine
- Show multiplicities on every association (e.g. Member 1 -- * Loan)

Task 3 — Sequence Diagram

Show the message flow for the 'Borrow Item' use case, including the system checking availability and updating the loan record.

Deliverable

Submit all three diagrams as a single PDF, built in draw.io, StarUML, or Lucidchart. Include a one-paragraph design rationale for each diagram.